



ADVANCED AI FULL COURSE

MASTER MACHINE LEARNING, DEEP LEARNING & BEYOND

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AI BASICS

1M+
Views

AI Basics Course link is in description box



ADVANCED AI FULL COURSE

MASTER MACHINE LEARNING, DEEP LEARNING & BEYOND

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Using AI Is Easy. Understanding Its Technology Is Power

AI Is More Than Tools: It's Algorithms, Models, and Intelligence



MODERN AI & DEEP
LEARNING ECOSYSTEM



MCP
(Model Context Protocol)



TRANSFORMER
ARCHITECTURE



LLM
(Large Language Models)



ATTENTION
MECHANISM

SELF-ATTENTION



RAG
(Retrieval-Augmented
Generation)



MULTIMODAL
MODELS



AI AGENTS



REINFORCEMENT
LEARNING (RL)



DIFFUSION
MODELS



EMBEDDINGS



TOKENIZATION



INFERENCE
ENGINE



FINE-TUNING



ChatGPT

Gemini



Grok



DALL-E



Sora

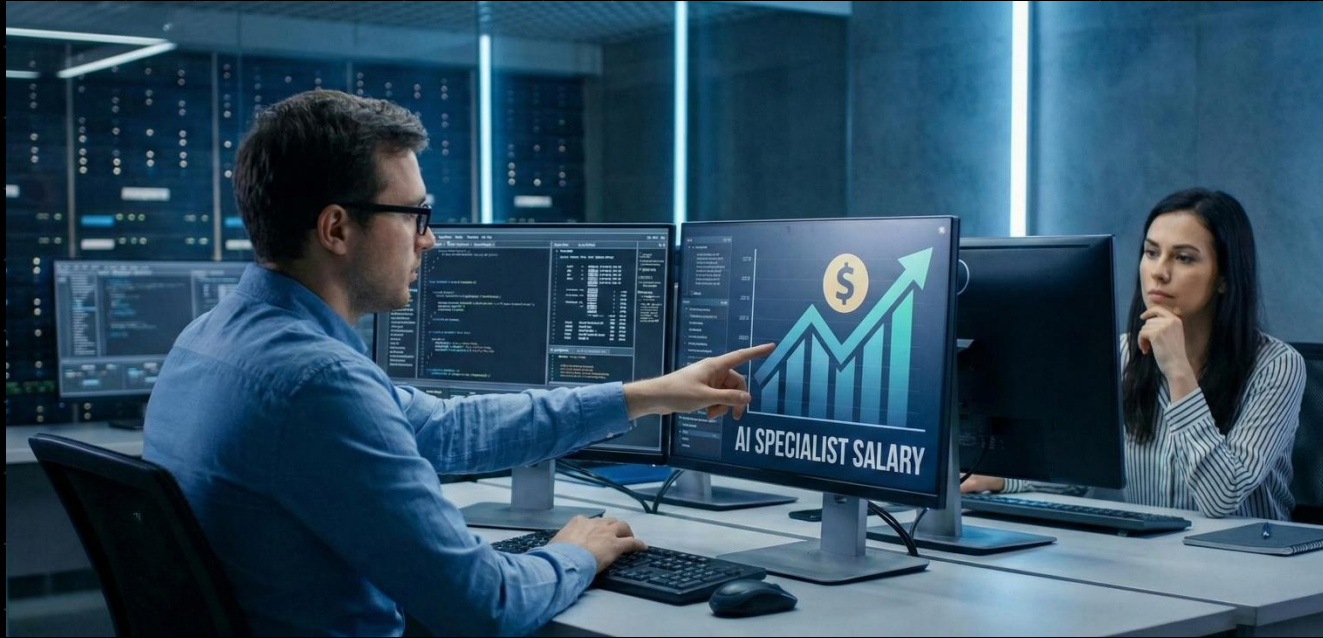


manus



DEVIN

Higher Salaries Come from Deeper AI Knowledge - Not Just Tools



AI KNOWLEDGE HIERARCHY & EVOLUTION



LEVEL 5: AUTONOMOUS AI & FUTURE TECH (Cutting-Edge)

- 26. AI Agents
- 27. Edge AI
- 28. Zero-Shot Learning
- 29. AGI (Artificial General Intelligence)
- 30. ASI (Artificial Super Intelligence)



LEVEL 4: INFORMATION RETRIEVAL & KNOWLEDGE SYSTEMS (Advanced+)

- 21. Vector Database
- 22. RAG (Retrieval Augmented Generation)
- 23. Knowledge Graphs
- 24. Explainable AI (XAI)
- 25. Federated Learning



LEVEL 3: AI MODELING & GENERATIVE AI (Advanced)

- 13. LLM (Large Language Model)
- 14. Fine-Tuning
- 15. LORA (Low-Rank Adaptation)
- 16. GAN (Generative Adversarial Network)
- 17. Diffusion Models
- 18. Multimodal AI
- 19. Vision-Language Models (VLMs)
- 20. Synthetic Data



LEVEL 2: MACHINE LEARNING & DEEP LEARNING CONCEPTS (Intermediate)

- 6. CNN (Convolutional Neural Network)
- 7. RNN (Recurrent Neural Network)
- 8. Transformers
- 9. Self-Supervised Learning
- 10. Reinforcement Learning (RL)
- 11. AutoML
- 12. Model Compression (Pruning & Quantization)

LEVEL 1: AI FOUNDATIONS (Basics)

- 1. Tokenization
- 2. Embeddings
- 3. Inference
- 4. Neural Network
- 5. MCP (Multi-Layer Perceptron)



What You Will Learn in This Course

1 LEVEL 1: AI FOUNDATIONS

- Tokenization
- Embeddings
- Inference
- Neural Network
- MCP (Model Context Protocol)

2 LEVEL 2: MACHINE LEARNING & DEEP LEARNING CONCEPTS

- CNN (Convolutional Neural Network)
- RNN (Recurrent Neural Network)
- Transformers
- Self-Supervised Learning
- Reinforcement Learning (RL)
- AutoML
- Model Compression (Pruning & Quantization)

What You Will Learn in This Course

3

LEVEL 3: AI MODELING & GENERATIVE AI

- LLM (Large Language Model)
- Fine-Tuning
- LORA (Low-Rank Adaptation)
- GAN (Generative Adversarial Network)
- Diffusion Models
- Multimodal AI
- Vision-Language Models (VLMs)
- Synthetic Data

4

LEVEL 4: INFORMATION RETRIEVAL & KNOWLEDGE SYSTEMS

- Vector Database
- RAG (Retrieval Augmented Generation)
- Knowledge Graphs
- Explainable AI (XAI)
- Federated Learning

What You Will Learn in This Course

5

LEVEL 5: AUTONOMOUS AI & FUTURE TECH (Cutting-Edge)

- AI Agent
- Edge AI
- Zero-Shot Learning
- AGI (Artificial General Intelligence)
- ASI (Artificial Super Intelligence)

Let's Begin the Advanced AI Journey

1

LEVEL 1: AI FOUNDATIONS

- Tokenization
- Embeddings
- Inference
- Neural Network
- MCP (Model Context Protocol)

LEVEL 1: AI FOUNDATIONS

1 Tokenization in AI

Tokenization is the process of breaking a long piece of text into smaller units called tokens so that an AI model can understand, analyze, and process the text.



ChatGPT

Gemini

Grok



LEVEL 1: AI FOUNDATIONS

1 Tokenization in AI

AI systems cannot read long sentences the way humans do.

They require text to be converted into small, structured pieces → tokens → which can later be converted into numbers.

Token kya hota hai?

Token ka simple matlab:

- ✓ Words ka small piece
- ✓ Ek pura word
- ✓ Ek character
- ✓ Ya word ke parts (subwords)



ChatGPT

Gemini

Grok



1 Tokenization in AI

Sentence:

"Artificial Intelligence is powerful."

Step-by-step tokenization:

A. Word-level tokenization

["Artificial", "Intelligence", "is", "powerful"]

B. Character-level tokenization

["A", "r", "t", "i", "l", "i", "c", "i", "a", "l", ...]

C. Subword-level tokenization (LLMs me mostly use hota hai)

["Art", "ificial", "Intelli", "gence", "is", "power", "ful"]

Isme AI complex words ko small meaningful pieces me todta hai. Kyuki agar kal ko koi naye word aa jaye jo model ne training me nahi dekha, tab bhi wo word ke sub-parts ko samajh sakta hai.



1 Tokenization in AI

Tokenization kyu zaroori hai?

Tokenization ke bina AI:

- ✗ Text ka structure nahi samajh payega
- ✗ Word boundaries nahi pehchan payega
- ✗ Similar words ka relationship track nahi kar sakta
- ✗ Efficient processing possible nahi hoti

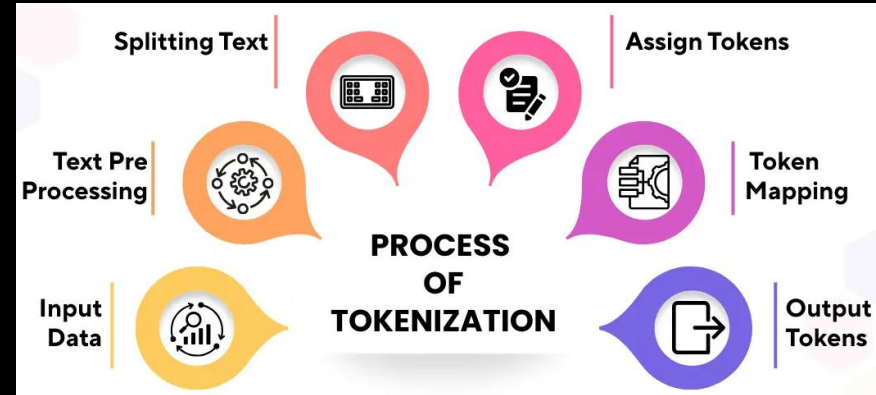
Tokenization AI ko yeh samajhne me help karta hai:

Sentence me kaunsi cheez important hai

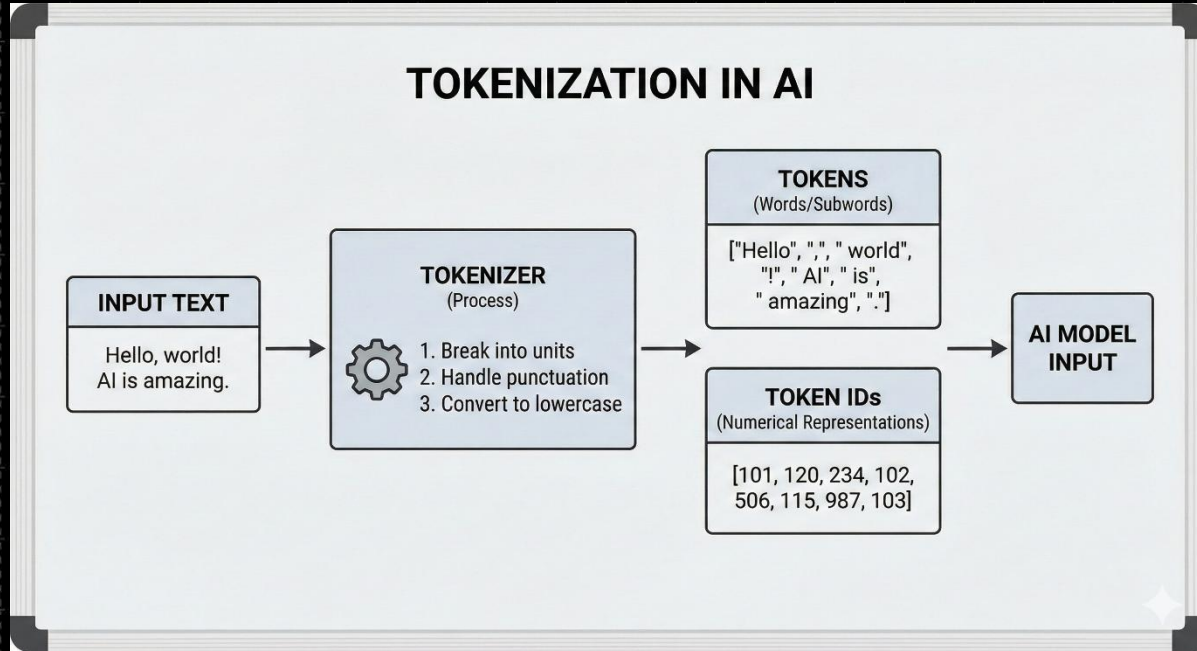
Kaunsa word kya meaning rakhta hai

Word ke similar parts kaise behave karte hain

Long text ko small manageable parts me kaise break kare



1 Tokenization in AI



AI models tokens ko numbers me kaise convert karte hai?

Har token ka ek unique ID (number) hota hai.

Example:

"Data" → 1432

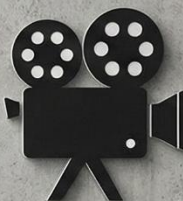
"Sci" → 987

"ence" → 654

"," → 29

Phir AI in numbers ko vectors me convert karta hai — ise bolte hain **Embedding**.

 **AUDIO** 

 **VIDEO** 

 **I M A G E** 

Image Tokenization

AI model image ko seedha ek photo ki tarah samajh nahi sakta.
AI ko image ko math form (numbers) me convert karna hota hai.
Ye numbers banane ke liye image ko tokens me todna padta hai.

Image Tokenization kaise hota hai?

Step 1: Image $\rightarrow$ Patches (Chhote squares)

Example:

Ek 256×256 pixel ki image ko 16×16 pixel ke patches me tod diya.

$256/16 = 16$ patches per row

Total patches: $16 \times 16 = 256$ patches

Ye chhote patches AI ke liye "words" ki tarah hote hain.

Step 2: Patches $\rightarrow$ Numbers (Flattening)

Har patch ko flatten karke ek vector banaya jata hai.

Step 3: Vector $\rightarrow$ Token

Ye vector embedding layer se pass hota hai aur token ban jata hai.

The Result: The image is now a sequence of vectors. To the AI, the image looks just like a sentence, where every 16×16 patch is equivalent to one "word."



Image Tokenization

Image Tokenization kaise hota hai?

Step 1: Image $\rightarrow$ Patches (Chhote squares)

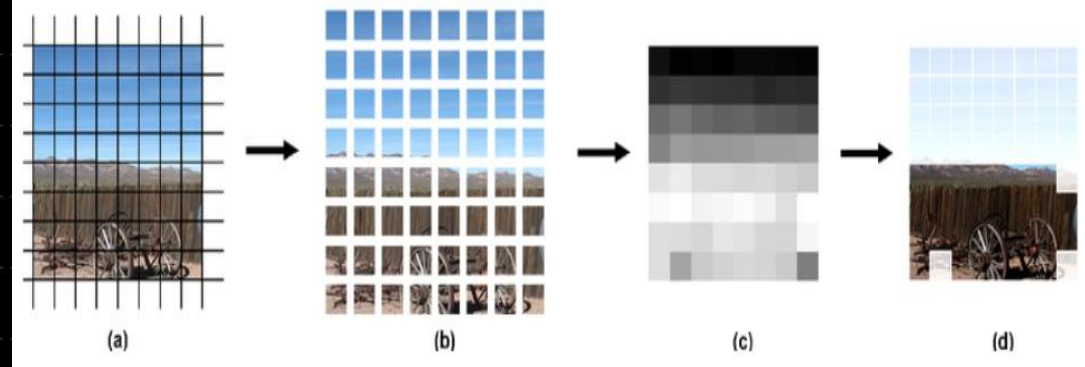
Example:

Ek 256×256 pixel ki image ko 16×16 pixel ke patches me tod diya.

Step 2: Patches $\rightarrow$ Numbers (Flattening)

Step 3: Vector $\rightarrow$ Token

The Result: The image is now a sequence of vectors. To the AI, the image looks just like a sentence, where every 16×16 patch is equivalent to one "word."



Advanced Method (VQ-VAE) - VQ-VAE stands for Vector Quantized Variational AutoEncoder.

For generating images, AI often uses a "Visual Codebook." It compares each patch to a dictionary of known visual patterns (e.g., "edge," "blue curve") and assigns it a specific ID number. The image becomes a list of IDs, like [452, 112, 889...]

"DALL-E 1" used this exact VQ-VAE method to learn how to generate images from text for the first time

Audio Tokenization

Audio ek continuous wave hoti hai.

AI continuous signal ko directly samajh nahi sakta, isliye isko small parts (frames) me todta hai — yehi hota hai audio tokenization.

★ Audio Tokenization kaise hota hai?

Step 1: Audio → Frames (10–25 ms segments)

1 second audio = approx 100–200 frames

Each frame = 10ms ya 20ms ka sound piece

Step 2: Frames → Spectrogram

Har frame ka frequency & amplitude calculate karke spectrogram banaya jata hai. Ye ek image jaisa hota hai (mel-spectrogram).

Step 3: Spectrogram → Audio Tokens

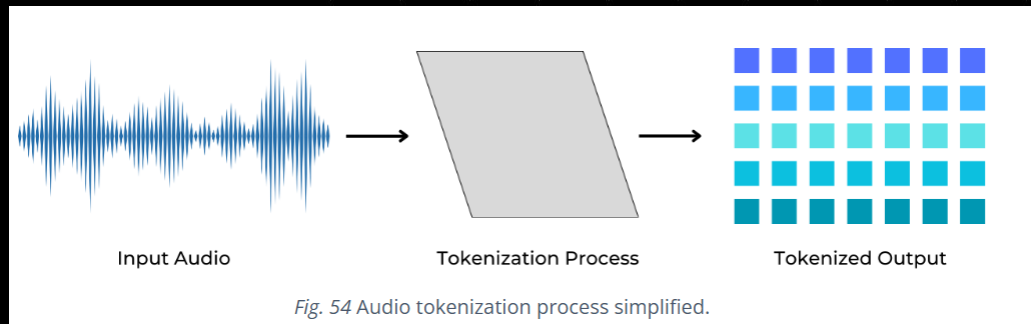
AI is spectrogram ko small grid patches me todkar tokens banata hai, bilkul image ki tarah.

★ Example

“I love data science.”

AI isko token me break karega:

Small audio frames → spectrogram patches → tokens → numbers → model.



Video = 📷 Images (frames) + 🔊 Audio + 🌀 Motion
 Isliye video tokenization 3 parts me hota hai.

★ Video Tokenization kaise hota hai?

Step 1: Video → Frames

Ek 1-second video me 24–60 frames hoti hain.

Har frame ek image hai → isko image tokenization se patches me todte hain.

Example:

1 sec, 30 fps → 30 images → each image 256 patches → total 7680 tokens

Step 2: Motion Tokenization

Frames ke beech difference (movement) detect hota hai:

Optical flow

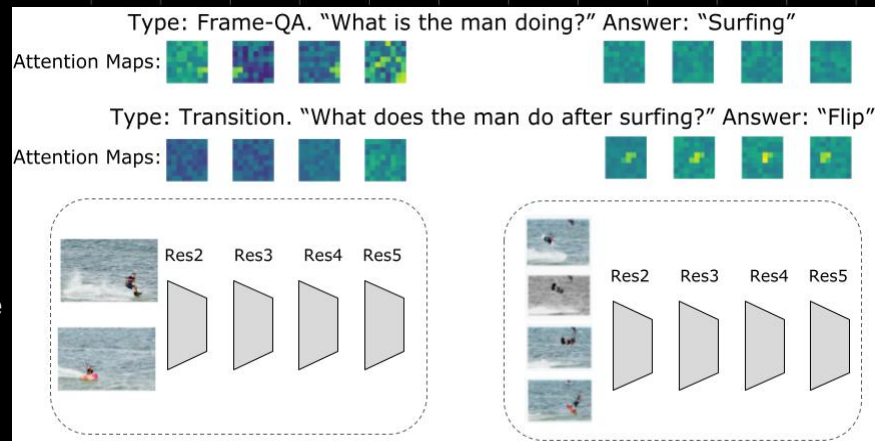
Motion vectors

Temporal patches

Ye motion tokens ban jaate hain.

Step 3: Audio Tokenization

Video ke audio part ka tokenization bhi hota hai using spectrograms

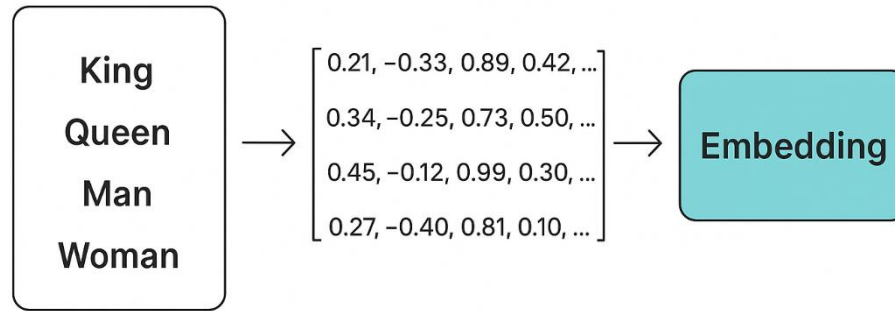


LEVEL 1: AI FOUNDATIONS

2 Embedding in AI

Agar simple shabdon mein kahein, toh Embedding kisi bhi data (text, image, audio) ko numbers ki ek list (vector) mein convert karne ka tareeka hai, taaki computer uske meaning ko samajh sake.

Embeddings in AI



Words convert to vectors

2

Embedding in AI

Embedding Kya Hai?

Imagine karo aapko computer ko samjhana hai ki "Apple" aur "Banana" similar hain (dono fruit hain), lekin "Apple" aur "Car" alag hain.

Agar aap sirf ID doge (e.g., Apple=1, Banana=2, Car=3), toh computer ko lagega 2 aur 3 (Banana aur Car) zyada close hain, jo ki galat hai.



Embedding har word ko ek lambi list of numbers (coordinates) mein convert kar deta hai, jise Vector kehte hain. Example:

Apple: [0.9, 0.1, 0.05, ...]

Banana: [0.88, 0.12, 0.04, ...]

Car: [0.1, 0.8, 0.9, ...]

(Apple ke numbers ke bahut paas)

(Bilkul alag numbers)

Isse machine ko pata chalta hai ki Apple aur Banana ka "meaning" paas-paas hai.

Yeh Kaam Kaise Karta Hai?

Embedding ko aap ek Map (Naksha) ki tarah samajh sakte ho.

Is map par har word ka ek address (coordinates) hota hai.

Jo words meaning mein same hote hain, wo map par paas-paas hote hain (Jaise: Happy aur Joy).

Jo words alag hote hain, wo door hote hain (Jaise: Happy aur Sad).

Vector: Numbers ki wo list jo data ko represent karti hai. Example: [0.12, -0.45, 0.88].

Dimensions: Vector kitna lamba hai. OpenAI ke models mein aksar 1536 dimensions (numbers) hote hain ek word/sentence ke liye. Jitne zyada dimensions, utni zyada details (meaning) capture hogi.

Cosine Similarity: Yeh ek formula hai jisse hum measure karte hain ki do vectors (embeddings) kitne close hain. Agar score 1 hai toh bilkul same, 0 hai toh koi connection nahi.

2

Embedding in AI

Embeddings ke Types

Mainly do tarah ki situations hoti hain:

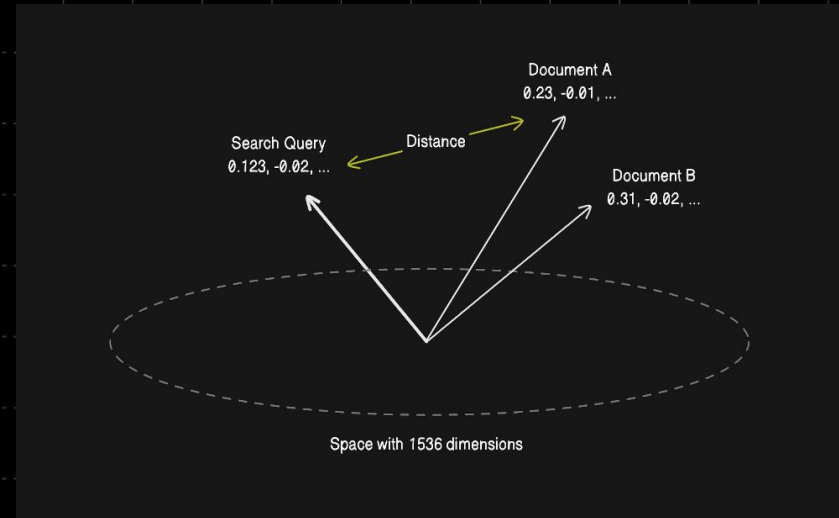
A. Word Embeddings (Purane Models) Isme har word ka ek fixed vector hota tha.

Problem: Word "Bank" ka matlab river bank bhi ho sakta hai aur money bank bhi. Purane models (jaise Word2Vec) confuse ho jate the kyunki unke paas "Bank" ke liye ek hi vector tha.

B. Contextual Embeddings (Modern Models - LLMs) Aajkal ke AI (jaise GPT-4, BERT) Contextual Embedding use karte hain.

Yeh poore sentence ko padhte hain. Agar sentence hai "I went to the bank to deposit money", toh yahan "bank" ka embedding financial institution wala banega.

Isliye aaj kal ke AI human-like samajh rakhte hain.



Real World Use Cases

Semantic Search (Google Search): Agar aap search karo "shoe for running", toh result mein "jogging sneakers" bhi aayega. Bhale hi word match nahi hua, par embedding ne pakad liya ki running aur jogging ka meaning same hai.

Recommendation Systems (Netflix/YouTube): Agar aapne ek sci-fi movie dekhi, toh system us movie ka embedding dekhega aur vector space mein uske paas wali doosri movies recommend karega.

RAG (Retrieval-Augmented Generation): Aajkal Custom Chatbots banane mein iska bahut use hota hai. Aap apne business ka data embeddings mein convert karke database (Vector Database) mein daal dete ho. Jab user sawal puchta hai, AI relevant data dhundh kar answer deta hai.

Sentiment Analysis: Review positive hai ya negative, yeh samajhne ke liye text ko numbers mein badla jata hai.

LEVEL 1: AI FOUNDATIONS

3

Inference in AI

AI me do main phases hote hain:



Training Phase

Model ko data + correct answers (labels) diye jaate hain
Model patterns seekhta hai

Example:

10,000 images → “ye dog hai”, “ye cat hai”
Model sikhta hai: dog kaisa dikhta hai



Inference Phase

Ab model seekhna band karta hai
Naye data par answer deta hai

Example:

New image → model bolta hai: “95% probability ye dog hai”



Real-world me hum mostly inference hi use karte hain.

LEVEL 1: AI FOUNDATIONS

3

Inference in AI

Inference ka simple real-life example



Doctor Example

Doctor ne years tak padhai ki → Training

Patient aaya → symptoms dekhe → diagnosis bataya

→ Inference



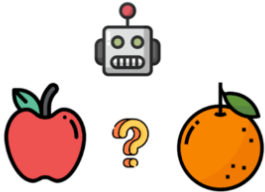
AI Example

Model ne lakhon examples dekhe → Training

User ne input diya → model ne output diya → Inference

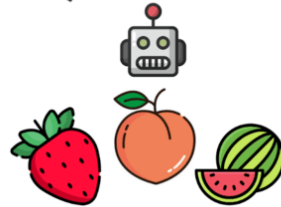
Training

Trained on Pre-defined Fruits



AI Inference

Guessing New and Unseen Fruits



LEVEL 1: AI FOUNDATIONS

4

Neural Network in AI

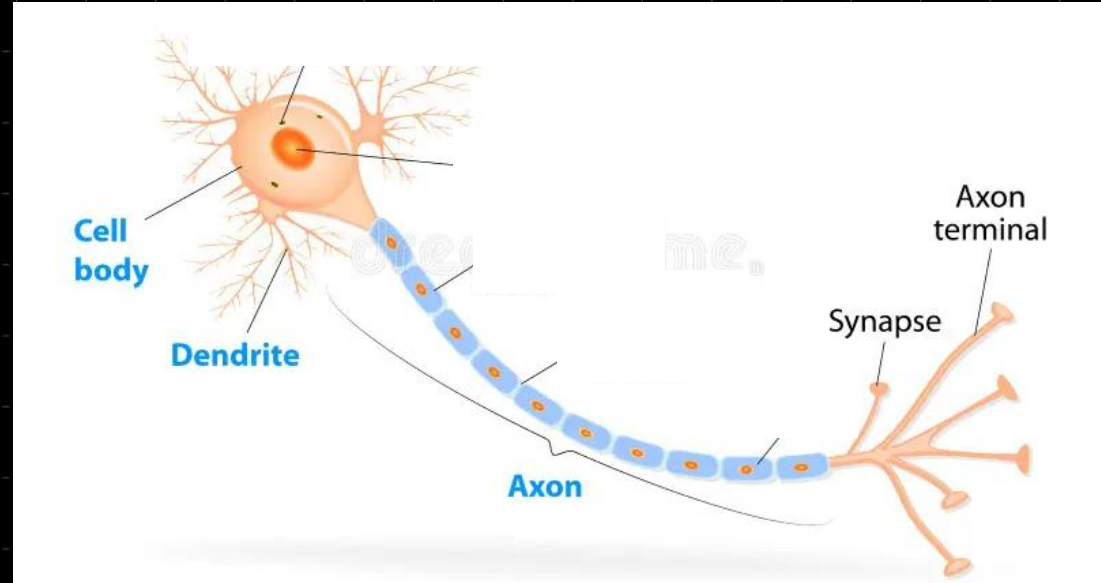
The Biological Neuron (Human Brain)
This is the physical cell in your head.

Dendrites (The Receivers): These are tree-like branches that receive chemical signals from other neurons. They are the "Input" collectors.

Cell Body / Nucleus (The Processor): This is the center of the cell. It collects all the signals coming from the dendrites and decides if the signal is strong enough to pass forward.

Axon (The Cable): If the signal is strong, it travels down this long tail. It is the transmission line.

Synapse (The Connection): This is the gap between the axon of one neuron and the dendrite of the next. The strength of this connection determines how much signal gets through.



LEVEL 1: AI FOUNDATIONS

4

Neural Network in AI

The Artificial Neuron (AI Model)

This is the mathematical code in the computer.

Inputs : These replace the Dendrites. This is the data entering the system (like pixels of an image).

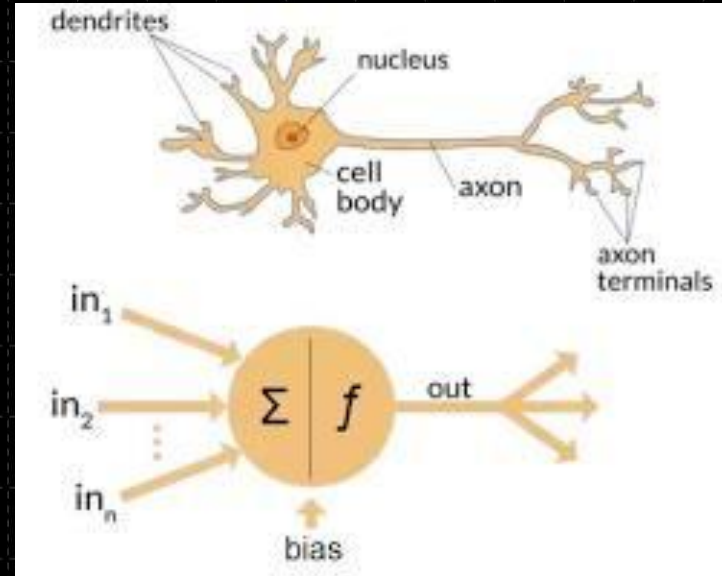
Weights : These replace the Synapse. Every input has a "weight" (importance). If a weight is high, that input is treated as very important.

Summation & Activation Function: These replace the Cell Body.

Summation : The computer adds up all the inputs multiplied by their weights.

Activation Function: This is a mathematical rule (like a threshold) that decides: "Is this number big enough to fire?" If yes, it passes the result forward.

Output : This replaces the Axon. It is the final result sent to the next layer of the network.



LEVEL 1: AI FOUNDATIONS

4

Neural Network in AI

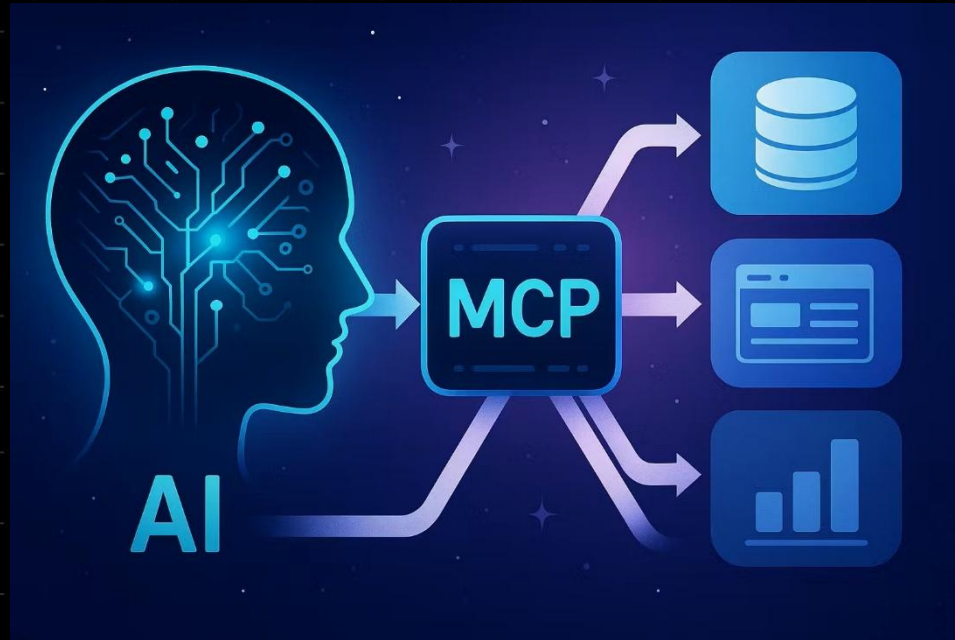
Human Brain (Biology)	AI Neural Network (Math)	Function (Kaam kya hai?)
Dendrites	Inputs (x)	Data receive karna (Receive signals).
Synapse	Weights (w)	Connection ki strength (How important is the signal?).
Cell Body (Soma)	Summation ($\sum$)	Saare signals ko jodna (Calculate total input).
Threshold / Firing	Activation Function	Decide karna ki signal aage bhejna hai ya nahi (Decision making).
Axon	Output (y)	Result ko aage transfer karna (Transmit result).

LEVEL 1: AI FOUNDATIONS

5 MCP (Model Context Protocol) in AI

Model Context Protocol (MCP) AI ki duniya mein ek naya aur revolutionary open standard hai. Ise simple bhaasha mein samjhein toh yeh AI models ke liye "USB port" ki tarah hai.

Jaise USB cable se aap mouse, keyboard, ya printer kisi bhi computer mein laga sakte hain, waise hi MCP ke through AI models kisi bhi data source ya tool se connect ho sakte hain.



LEVEL 1: AI FOUNDATIONS



MCP (Model Context Protocol) in AI

MCP (Model Context Protocol) ek open standard hai jise Anthropic ne introduce kiya hai. Iska main kaam hai AI assistants (jaise Claude, ChatGPT) ko system ke data (files, servers, databases) ke saath secure aur standardized tarike se connect karna.

Example: Pehle agar aapko Claude ko Google Drive, Slack, aur GitHub se connect karna hota, toh developers ko har tool ke liye alag-alag integration likhna padta tha.

MCP se pehle: Har AI model X Har Tool = Bahut saare connections (Messy).

MCP ke baad: Ek standard protocol jo sab jagah chalega.

The USB Analogy: Sochiye MCP "USB-C" hai.

AI Model aapka Laptop hai.

External Tool (Google Drive/Slack) aapka Printer ya Hard Drive hai.

MCP wo cable hai jo bina kisi naye driver ke inhe connect kar deti hai.

LEVEL 1: AI FOUNDATIONS

5

MCP (Model Context Protocol) in AI

MCP Kaise Kaam Karta Hai?

MCP architecture mein mainly teen parts hote hain. Isko samajhna zaroori hai:

MCP Host (The App): Yeh wo application hai jisme AI model chalta hai. Example ke liye Claude Desktop App ya Cursor IDE. Yeh host decide karta hai ki user kis tool ko access kar sakta hai.

MCP Client (The Brain): Yeh Host ke andar hota hai. Yeh AI model aur MCP Server ke beech baat karwata hai.

MCP Server (The Data Provider): Yeh sabse important part hai. Yeh ek lightweight program hota hai jo aapke data (jaise local files, PostgreSQL database, ya Google Drive) ko expose karta hai taaki AI usse padh sake.

Process Flow: User Request → MCP Host → MCP Client → MCP Server → Data Fetch → AI Response

LEVEL 2 :Advanced Machine Learning & Deep Learning

6 Convolutional Neural Networks in AI

CNN (Convolutional Neural Networks) AI aur Deep Learning ka ek bahut hi important part hai. Agar simple shabdon mein kahein, toh CNN computer ki "aankhein" (eyes) hain.

Jab bhi computer ko koi photo, video, ya visual data dekh kar samajhna hota hai (jaise face unlock, self-driving cars), toh wahan CNN ka use hota hai.



LEVEL 2 :Advanced Machine Learning & Deep Learning

6

Convolutional Neural Networks in AI



CNN Kya Hai?

CNN ka full form hai Convolutional Neural Network. Yeh ek tarah ka Deep Learning algorithm hai jo images ko input ke taur par leta hai, usme patterns (features) ko pehchanta hai, aur batata hai ki wo image kis cheez ki hai.

Analogy: Sochiye aap ek Gulab (Rose) ko kaise pehchante hain?

Pehle aap dekhte hain ki petals ka shape kaisa hai (curved lines).

Fir aap color dekhte hain (Red/Pink).

Fir aap stem (tana) aur kaante dekhte hain.

Aakhir mein, aapka dimaag decision leta hai: "Yeh ek Gulab hai."

CNN bhi bilkul aise hi kaam karta hai. Wo puri image ko ek baar mein nahi ratta maarta, balki use tukdon (layers) mein todkar patterns dhundta hai.

LEVEL 2 :Advanced Machine Learning & Deep Learning

6 Convolutional Neural Networks in AI

Convolution: Features Dhundna. Image scan karke lines/shapes nikalta hai.

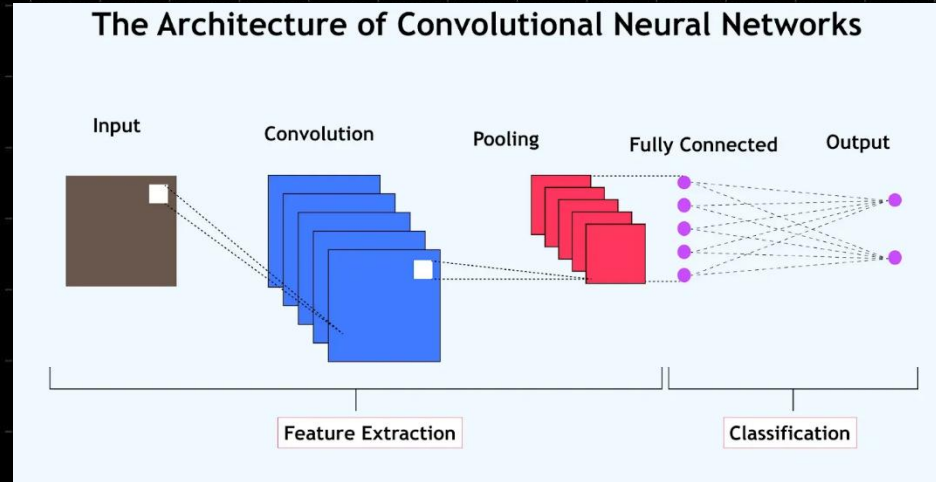
ReLU: Activation. Data ko clean karta hai (Negative values hataata hai).

Pooling: Size Chhota Karna. Image ko compress karta hai taaki fast chale.

Flattening: Seedha Karna. 2D image data ko 1D list (line) banata hai.

Fully Connected: Decision Lena. Patterns jodkar sochta hai ki yeh kya hai.

Output: Result Dena



LEVEL 2 :Advanced Machine Learning & Deep Learning

7

Recurrent Neural Network in AI

RNN (Recurrent Neural Networks) AI ka wo hissa hai jo unn cheezon ko samajhne ke liye bana hai jahan Order (Sequence/Kram) aur Time (Waqt) matter karta hai.

Agar CNN computer ki "Aankhein" (Eyes) hain, toh RNN computer ki "Yaadasht" (Short-term Memory) hai.

RNN Kya Hai? (What is RNN?)

RNN (Recurrent Neural Network) ek aisa Deep Learning network hai jo purani information ko yaad rakhta hai taaki aage aane wali information ko samajh sake

Best Example:

Normal Network: Har shabd ko alag dekhega: "I", "Love", "India".

RNN: "I" dekha → Memory mein rakha. "Love" dekha → "I" ke saath connect kiya. "India" dekha → Teeno ko connect karke samjha ki baat desh-bhakti ki ho rahi hai.



LEVEL 2 :Advanced Machine Learning & Deep Learning

7

Recurrent Neural Network in AI

RNN Kaise Kaam Karta Hai? (The Mechanism)

RNN ke kaam karne ka tarika ek Loop jaisa hota hai.

Iske structure mein ek khaas cheez hoti hai jise "Hidden State" kehte hain.

Input : Naya data jo abhi aaya hai (Current word/input).

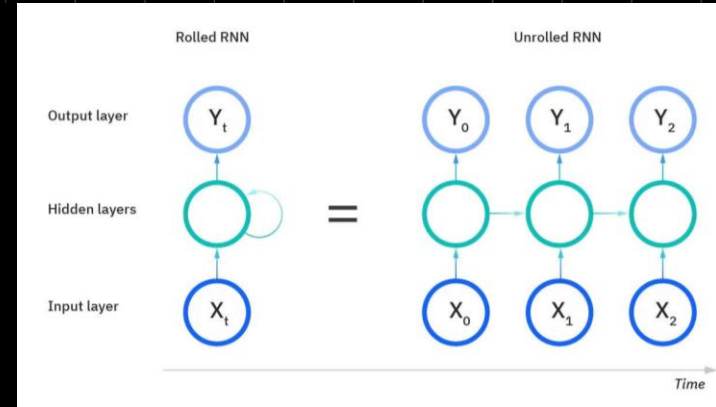
Hidden State : Yeh RNN ki Memory hai. Yeh pichle step ka output hota hai jo current step mein wapas feed kiya jata hai.

The Loop: RNN naye input ko purani memory ke saath combine karta hai aur ek nayi memory create karta hai.

Equation ka Logic (Simple Bhaasha mein):

Aaj Ka Faisla = (Abhi Ki Khabar) + (Pichla Experience)

Current Output = Current Input + Previous Hidden State



LEVEL 2 :Advanced Machine Learning & Deep Learning

7

Recurrent Neural Network in AI

🌟 RNN Kahan Use Hota Hai? (Applications)

Jahan bhi data ek sequence mein ho, wahan RNN use hota hai:

Google Translate (Machine Translation): English sentence pura sunkar use Hindi mein convert karna. (Grammar ka order change hota hai, isliye memory chahiye).

Speech Recognition (Siri/Alexa): Audio waves lagatar aati hain. "Hey" ke baad "Siri" aane ki probability RNN hi calculate karta hai.

Predicting Next Word: Jab aap mobile par type karte hain aur keyboard suggest karta hai agla word, wo RNN (ya uske advanced versions) kar rahe hote hain.

Stock Market Prediction: Aaj ka share price pichle 10 din ke trend par depend karta hai. Sequence matter karta hai.

LEVEL 2 :Advanced Machine Learning & Deep Learning

8

Transformers in AI

Transformers AI ki duniya ka sabse bada "Game Changer" hai. Aajkal aap jo bhi powerful AI dekh rahe hain (jaise ChatGPT, Gemini, Claude), wo sab Transformers architecture par hi bane hain.

Agar RNN "Short-term Memory" tha, toh Transformer ek "Super Computer Brain" hai jo puri kitaab ek saath padh sakta hai aur sab yaad rakh sakta hai.

Ise 2017 mein Google ke researchers ne introduce kiya tha ek famous paper mein jiska naam tha: "Attention Is All You Need".



LEVEL 2 :Advanced Machine Learning & Deep Learning

8

Transformers in AI

🤖 Transformer Kya Hai? (What is a Transformer?)

Transformer ek Deep Learning model hai jo Sequence-to-Sequence tasks (jaise Translation, Text Summarization) ke liye banaya gaya tha.

Lekin isne purane models (RNN/LSTM) ki sabse badi kami door kar di: Sequential Processing.

RNN: Ek-ek shabd padhta tha (One by one). Isliye slow tha aur lambi baatein bhool jata tha.

Transformer: Pura sentence ek saath (Parallel) dekhta hai. Isliye yeh bahut fast hai aur door-door ke words ka rishta bhi samajh leta hai.

💡 Iska Secret Sauce: "Self-Attention" Mechanism

Transformer ki sabse khaas cheez hai "Self-Attention". Iska matlab hai sentence ke har word ka dusre words se connection check karna.

Example se samjhein: Sentence: "The animal didn't cross the street because it was too tired."

Jab insaan isko padhta hai, wo jaanta hai ki "it" ka matlab "animal" hai, "street" nahi.

Purane models (RNN) confuse ho jate the kyunki "it" aur "animal" ke beech doori hai.

Transformer (Attention) har word ko highlight karta hai. Jab wo "it" par pahunchta hai, toh wo piche mudkar dekhta hai aur "animal" par sabse zyada attention (dhyan) deta hai.

LEVEL 2 :Advanced Machine Learning & Deep Learning

8

Transformers in AI

Transformer Ka Structure (Architecture)

Transformer mainly do hisson mein banta hota hai: Encoder aur Decoder.

1. Encoder (The Reader / Samajhne wala) 📖

Iska kaam hai input data (English sentence) ko lena aur use samajhna. Yeh data ke features aur meaning nikalta hai.

Yeh input words ko numbers (vectors) mein convert karta hai.

Phir "Self-Attention" use karke context samajhta hai.

2. Decoder (The Writer / Banane wala) 🙌

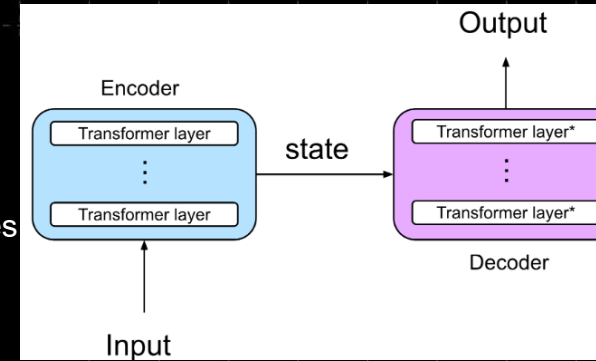
Iska kaam hai Encoder ki samajh ko lekar output generate karna (Hindi sentence ya Answer).

Yeh predictions karta hai ki agla word kya hona chahiye.

Note:

BERT (Google Search mein use hota hai) sirf Encoder use karta hai (sirf samajhne ke liye).

GPT (ChatGPT) mainly Decoder use karta hai (text generate karne ke liye).



LEVEL 2 :Advanced Machine Learning & Deep Learning

9

Self Supervised Learning in AI

Self-Supervised Learning (SSL) AI ki duniya mein ek revolutionary concept hai. Agar simple shabdon mein kahein, toh ye wo technique hai jahan AI model bina kisi human help (labels) ke khud hi seekhta hai.

Isse pehle ki hum deep dive karein, ek simple analogy samajhte hain:

Analogy: Imagine karo ek student hai jo library mein baitha hai. Uske paas karodon books hain, lekin koi teacher nahi hai jo bataye ki "Ye history hai" ya "Ye science hai".

Wo student kya karta hai? Wo sentences ke aadhe words ko dhak leta hai aur guess karta hai ki wahan kya likha hoga. Agar wo sahi guess karta hai, matlab use language samajh aa rahi hai. Yehi Self-Supervised Learning hai.



LEVEL 2 :Advanced Machine Learning & Deep Learning

9

Self Supervised Learning in AI

Self-Supervised Learning Kya Hai?

Traditionally, AI do tarah se seekhta tha:

Supervised Learning: Jahan hum model ko batate hain (Label dete hain), e.g., "Ye photo billi ki hai."

Unsupervised Learning: Jahan hum data de dete hain aur model patterns dhundhta hai (Clustering).

Self-Supervised Learning in dono ke beech ka "Sweet Spot" hai. Isme hum Unlabeled Data (jiska koi answer key nahi hai) use karte hain, lekin model data ke andar se hi labels generate kar leta hai.

Core Mantra: The Data is its own Label. (Data khud hi apna answer hai).

9

Self Supervised Learning in AI

Ye Kaam Kaise Karta Hai? (Mechanism)

SSL main do steps mein kaam karta hai. Is process ko aksar "Pre-training" aur "Fine-tuning" kaha jata hai.

Step 1: Pretext Task (The Learning Phase)

Model ko ek fake problem di jati hai solve karne ke liye, taaki wo data ke structure ko samajh sake. Isse "Pretext Task" kehte hain.

Examples of Pretext Tasks:

- Fill in the blanks (NLP):

- Input: "The sky is [MASK]."
- Model ka kaam: "[MASK]" ki jagah "blue" predict karna.
- *Result:* Is process se model grammar, context aur world knowledge seekh leta hai.

- Jigsaw Puzzle (Computer Vision):

- Input: Ek image ke 9 tukde kar diye aur unhe mix kar diya.
- Model ka kaam: Un tukdon ko wapas sahi order mein lagana.
- *Result:* Model ko samajh aata hai ki object kaisa dikhta hai (e.g., pair niche hote hain, sar upar).

Step 2: Downstream Task (The Application Phase)

Jab model "Pretext Task" se data ko samajh leta hai (Pre-trained ho jata hai), tab hum use kisi specific kaam (Real task) ke liye use karte hain.

- Ab hum thoda sa labeled data use karke use Fine-tune karte hain.

- Kyunki model pehle se intelligent ho chuka hai, wo bohot kam data mein bhi nayi cheez seekh leta hai.

LEVEL 2 :Advanced Machine Learning & Deep Learning

10

Reinforcement Learning in AI

Reinforcement Learning (RL) AI sikhne ka wo tareeka hai jo bilkul waise hi kaam karta hai jaise ek chhota bacha seekhta hai—Galtiyan karke aur unse seekh kar (Trial and Error).

Ye wo learning hai jahan koi teacher nahi hota jo answer bataye. Balki, agar aap sahi kaam karte ho toh Inaam (Reward) milta hai, aur galat karte ho toh Saza (Penalty) milti hai. Bas isi reward ko maximize karne ke chakkar mein AI intelligent ban jata hai.



LEVEL 2 :Advanced Machine Learning & Deep Learning

10

Reinforcement Learning in AI

RL ko samajhne ke liye ye 5 terms samajhna zaroori hai:

Agent (The Learner): Wo AI model jo seekh raha hai (e.g., Mario game khelne wala bot).

Environment (The World): Wo jagah jahan Agent kaam kar raha hai (e.g., Video game ka level ya Chess board).

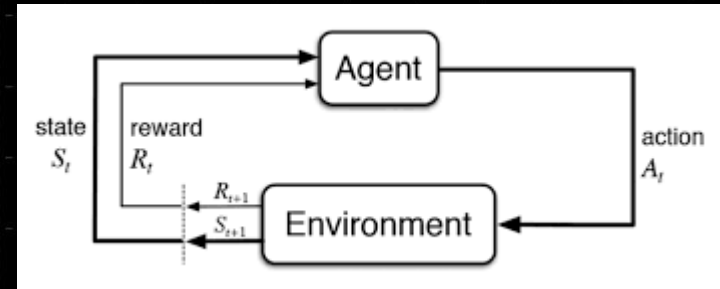
Action (The Move): Agent kya kar sakta hai (e.g., Left jana, Right jana, Jump karna).

Reward (The Feedback): Action ka result.

Positive (+10 points): Accha kaam kiya.

Negative (-5 points): Galti ki (e.g., aag mein gir gaya).

State (The Situation): Current situation kya hai (e.g., Mario abhi kahan khada hai).



10

Reinforcement Learning in AI

Ye Kaam Kaise Karta hai? (The Loop)

Ye ek continuous cycle hai jo chalti rehti hai:

- 1.Observation: Agent dekhta hai ki wo abhi kahan hai (State).
- 2.Decision: Wo decide karta hai ki kya karna hai (Action).
- 3.Feedback: Environment use batata hai ki ye action kaisa tha.
 1. Agar accha tha -> Reward.
 2. Agar bura tha -> Penalty.
- 4.Update: Agent apne dimaag (Policy) mein note kar leta hai ki "Is situation mein ye action lene se fayda hua ya nuksan".
- 5.Repeat: Ye process lakhon baar repeat hoti hai jab tak Agent perfect nahi ho jata.

11

Auto ML in AI

AutoML (Automated Machine Learning) ek aisi technology hai jo Machine Learning (ML) models banane ke process ko automate karti hai. Iska main goal ye hai ki non-experts bhi easily powerful AI models bana sakein aur experts ka time repetitive tasks se bache.

Analogy: Maan lo aapko ek cake banana hai.

Traditional ML: Aapko khud decide karna padega ki kitna aata lena hai, kitni cheeni, kitne degree par bake karna hai, aur kaunsa oven use karna hai. Agar galti hui, toh dobara try karna padega. (Ye experts karte hain).

AutoML: Aap bas ingredients (Data) machine mein daal dete ho, aur machine khud decide karti hai ki best recipe kya hogi, temperature kya hoga, aur perfect cake (Model) bana kar de deti hai.

11

Auto ML in AI

AutoML Kya Hai?

Traditionally, ek ML model banane ke liye ek **Data Scientist** ko bohot saare steps manually karne padte hain:

- 1.Data ko saaf karna (Cleaning).
- 2.Features choose karna (Feature Engineering).
- 3.Sahi algorithm select karna (e.g., Random Forest vs Neural Network).
- 4.Settings (Hyperparameters) ko tune karna.

Ye kaam bohot time-consuming aur mushkil hota hai. **AutoML** in sabhi steps ko (ya inmein se kuch ko) automate kar deta hai. Aap bas apna data upload karte ho, aur AutoML background mein hazaron combinations try karke aapko **Best Model** de deta hai.

11

Model Compression (Pruning & Quantization) in AI

Model Compression AI models ko "Diet" par bhejne jaisa hai. Aajkal ke AI models (jaise GPT, BERT) bohot "bhaari" (bulky) hote hain. Inke parameters billions mein hote hain aur size GBs mein hota hai. Inhe chalane ke liye mehenge GPUs aur servers lagte hain.

Agar humein in models ko Mobile Phones, Smart Watches, ya IoT devices par chalana hai (jahan memory aur battery kam hoti hai), toh humein model ka size kam karna padta hai bina uski intelligence (accuracy) khoye. Isi process ko Model Compression kehte hain.

Iske do sabse main tareeke hain: Pruning aur Quantization.



11

Model Compression (Pruning & Quantization) in AI

Pruning (Kaat-Chaat Karna)

Concept: Unn connections ko kaat dena jo zaroori nahi hain.

Neural Networks mein karodon connections (weights) hote hain. Training ke baad dekha gaya hai ki bohot saare connections ki value 0 ke paas hoti hai, yaani wo final result mein koi khaas yogdaan nahi de rahe.

Pruning ka matlab hai un bekar connections ko hata dena.

Analogy: Imagine karo ek bada ghana ped (tree) hai. Agar aap uski sookhi aur bekar tahniyan (branches) kaat doge, toh ped ka shape wahi rahega, lekin wo halka aur saaf ho jayega.



12

Model Compression (Pruning & Quantization) in AI

Quantization (Precision Kam Karna)

Concept: Numbers ki detail (resolution) ko kam karke memory bachana.

Computers mein normally AI models ke weights 32-bit Floating Point (FP32) mein store hote hain.

Example: 3.14159265 (Ye bohot jagah leta hai memory mein).

Quantization mein hum in numbers ko 8-bit Integers (INT8) mein convert kar dete hain.

Example: 3.14 ya sirf 3.

Analogy: Imagine karo aap ek 4K (Ultra HD) movie dekh rahe ho. File size 10GB hai. Agar aap use 720p (HD) mein convert kar do, toh file size 1GB reh jayega. Picture quality thodi kam hogi, lekin movie samajh mein aa jayegi. FP32 = 4K Movie | INT8 = 720p Movie

AI KNOWLEDGE HIERARCHY & EVOLUTION



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- 27. Edge AI
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LEVEL 4: INFORMATION RETRIEVAL & KNOWLEDGE SYSTEMS (Advanced+)

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LEVEL 3: AI MODELING & GENERATIVE AI (Advanced)

- 13. LLM (Large Language Model)
- 14. Fine-Tuning
- 15. LORA (Low-Rank Adaptation)
- 16. GAN (Generative Adversarial Network)
- 17. Diffusion Models
- 18. Multimodal AI
- 19. Vision-Language Models (VLMs)
- 20. Synthetic Data



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- 7. RNN (Recurrent Neural Network)
- 8. Transformers
- 9. Self-Supervised Learning
- 10. Reinforcement Learning (RL)
- 11. AutoML
- 12. Model Compression (Pruning & Quantization)

LEVEL 1: AI FOUNDATIONS (Basics)

- 1. Tokenization
- 2. Embeddings
- 3. Inference
- 4. Neural Network
- 5. MCP (Multi-Layer Perceptron)



What You Will Learn in This Course

3

LEVEL 3: AI MODELING & GENERATIVE AI

- LLM (Large Language Model)
- Fine-Tuning
- LORA (Low-Rank Adaptation)
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- Vision-Language Models (VLMs)
- Synthetic Data

13

Large Language Model in AI

LLM (Large Language Model) aaj ki AI revolution ka sabse bada hero hai. Agar aapne ChatGPT, Gemini, ya Claude use kiya hai, toh aap technically ek LLM se hi baat kar rahe the.

Simple shabdon mein kahein toh:

Analogy: Maan lijiye ek aisa insaan hai jisne internet ki **saari kitabein, articles, aur websites** padh li hain. Use har sawal ka jawab pata hai aur wo kisi bhi bhasha mein baat kar sakta hai. Bas yahi **LLM** hai.



13

Large Language Model in AI

LLM koi jaadu nahi hai, ye **Statistics aur Probability** par chalta hai. Iska main kaam bas ek hi hai: **"Agla shabd (Next Word) guess karna."**

Sochiye: Agar main kahun: *"Aaj mausam bohot _____"* Aapka dimaag turant bolega *"suhana hai"* ya *"garam hai"*. LLM bhi yahi karta hai. Wo pichle words ko dekh kar calculate karta hai ki agla shabd kya hona chahiye jo sabse sahi fit ho.

Lekin ye itna simple "Autocorrect" nahi hai. Ye poore paragraphs aur concepts ko connect kar sakta hai kyunki ye **Transformer Architecture** par bana hota hai.

Transformer Architecture (The Brain): 2017 mein Google ne 'Transformer' concept nikala tha. Isse pehle AI bhool jata tha ki paragraph ke shuru mein kya likha tha. Transformer mein **"Attention Mechanism"** hota hai, jo poore sentence mein *important words* par focus karta hai, chahe sentence kitna bhi lamba kyun na ho.

14

Fine Tuning in AI

Fine-Tuning AI ki duniya mein wahi process hai jo ek Student ko Specialist doctor banane ka hota hai. Agar simple shabdon mein kahein:

"Ek aise AI model ko jo pehle se kaafi kuch jaanta hai (Pre-trained), use kisi ek khaas kaam (specific task) ke liye expert banana hi Fine-Tuning hai."



The Best Analogy: MBBS Student vs. Heart Surgeon

Fine-Tuning ko samajhne ke liye ye example sabse best hai:

1.Pre-Trained Model (MBBS Student):

1. Ek student ne MBBS kiya. Use dawaiyon, anatomy, aur bimariyon ka **general knowledge** hai. Wo thoda bohot sab kuch janta hai.
2. *AI Context:* Ye **GPT-4** ya **BERT** jaisa bada model hai jisne internet ka saara data padh rakha hai.

2.Fine-Tuning (Specialization):

1. Ab wo student "Heart Surgery" ki special training leta hai. Wo ab history ya geography nahi padhta, sirf **Heart (Dil)** ke baare mein gehrai se padhta hai.
2. *AI Context:* Hum us AI model ko medical data dete hain taaki wo poetry likhna bhale hi bhool jaye, par **bimari pakadne mein expert** ho jaye.

LEVEL 3 : AI Modelling & Generative AI Advanced

15

LORA – Low Rank Adaption in AI

LoRA (Low-Rank Adaptation) AI fine-tuning ki duniya mein ek "**Smart Hack**" (Jugaad) hai.

Isse pehle humne "Fine-Tuning" samjha tha. Lekin traditional Fine-Tuning mein ek bohot badi problem hai:

Cost aur Hardware. Agar aapko GPT-3 ya Llama-3 jaise huge models ko train karna hai, toh aapko supercomputers chahiye. **LoRA ne is problem ko solve kar diya.**

LoRA ki wajah se aaj ek normal gaming laptop par bhi bade AI models ko train karna possible ho gaya hai.

15

LORA – Low Rank Adaption in AI

LoRA = Low-Rank Adaptation.

Iska main logic ye hai: Bade models mein jab hum nayi cheez sikhate hain, toh humein saare billion parameters ko change karne ki zaroorat nahi hoti. Actual changes bohot thode (Low Rank) hote hain.

Analogy (The Textbook Example):

- Pre-trained Model:** Ek moti College ki Textbook hai.
- Full Fine-Tuning:** Aap puri kitaab ke har page ko mita kar dobara likh rahe ho. (Very heavy).
- LoRA:** Aapne kitaab ko touch bhi nahi kiya. Aapne bas important pages par **Sticky Notes** chipka diye jisme naye notes likhe hain.
- Jab aap padhoge (Inference), toh aap Textbook + Sticky Notes dono ko mila kar padhoge.

16

GAN – Generative Adversarial Network

GAN (Generative Adversarial Network) AI ki duniya ka sabse creative aur fascinating concept hai. Isne AI ko sirf "Data Analyze" karne se "Data Create" karne ki power di hai.

Agar ek line mein samjhein:
"Ye do AI networks ki ladai (fight) hai, jisme ladte-ladte dono expert ban jate hain aur naya content (Images, Audio, Video) banana seekh lete hain."

Isse 2014 mein **Ian Goodfellow** ne banaya tha.

LEVEL 3 : AI Modelling & Generative AI Advanced

16

GAN – Generative Adversarial Network

Simple Analogy: Chor aur Police (Thief vs. Police)

GAN ko samajhne ke liye sabse famous example ye hai:

1.The Counterfeiter (Nakli Note Banane Wala Chor):

1. Iska kaam hai nakli note banana aur koshish karna ki wo asli lage.
2. Shuru mein wo bekar note banata hai.

2.The Police (Inspector):

1. Iska kaam hai note ko check karna aur batana ki ye "Asli" hai ya "Nakli".

Process:

- Chor note banata hai -> Police pakad leti hai ("Ye nakli hai!").
- Chor apni galti sudharta hai aur behtar note banata hai.
- Police bhi apni checking strict karti hai.
- Ye cycle chalti rehti hai.
- End Result:** Ek time aisa aata hai jab Chor itna perfect nakli note banata hai ki Police bhi fark nahi bata paati. Bas yahi **GAN** hai.

LEVEL 3 : AI Modelling & Generative AI Advanced

16

GAN – Generative Adversarial Network

GAN ke 2 Main Parts (Architecture)

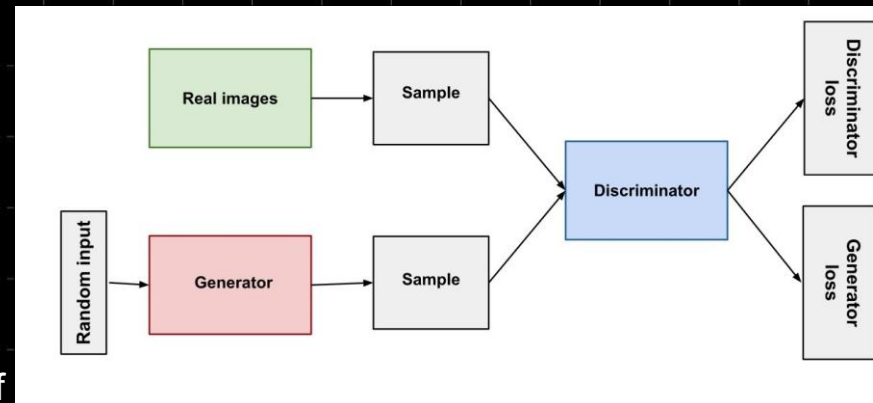
Technically, GAN mein do alag-alag Neural Networks hote hain jo ek dusre ke khilaf kaam karte hain (Adversarial matlab Virodhi):

A. Generator (The Artist/Faker)

- **Kaam:** Naya nakli data create karna.
- **Input:** Ye random numbers (Noise) leta hai.
- **Goal:** Aisi image banana jo Discriminator ko bewakoof bana sake.

B. Discriminator (The Critic/Judge)

- **Kaam:** Pehchanna ki data asli hai ya nakli.
- **Input:** Iske paas Real Images (Training Data) aur Generator ki Fake Images dono aati hain.
- **Goal:** Sahi classify karna (Real = 1, Fake = 0).



17

Diffusion Models

Diffusion Models aajkal ke **Generative AI** (Image generation) ka sabse powerful engine hai.

Agar aapne **MidJourney**, **DALL-E 3**, ya **Stable Diffusion** ka naam suna hai ya use kiya hai, toh samajh lijiye wo sab **Diffusion Models** par hi chalte hain.

Simple shabdon mein:

"Ye wo AI hai jo 'Shor' (Noise) mein se 'Tasveer' (Image) dhundh nikalta hai."



LEVEL 3 : AI Modelling & Generative AI Advanced

17

Diffusion Models

Simple Analogy: Dhundh se Saaf Tasveer (From Fog to Clarity)

Imagine kariye aapke paas ek saaf photo hai.

1. Aap uspar dhire-dhire **dhool (dust)** dalna shuru karte hain.

2. Thodi der baad wo photo dhundhli ho jayegi.

3. End mein, wo photo dikhna band ho jayegi, sirf **random noise (TV ki jhilmil)** bachegi.

Ab imagine kariye ek aisa **Artist (AI)** jise humne ye sikhaya hai ki **"Is dhool ko wapas kaise hatana hai?"**

Agar wo artist us random dhool (Noise) ko step-by-step saaf kare, toh wapas ek nayi sundar tasveer ban jayegi.

Bas yahi Diffusion Model hai.

17

Diffusion Models

Ye Kaam Kaise Karta Hai? (The 2-Step Process)

Diffusion models do main steps mein kaam karte hain: Forward aur Reverse.

Step 1: Forward Diffusion (Destruction - Barbad Karna)

- Hum training data (e.g., Cat ki photo) lete hain.
- Hum uspe thoda-thoda karke Noise (Random pixels) add karte jate hain.
- Result: Ek point par photo puri tarah se Pure Noise ban jati hai. Isme ab kuch nahi dikh raha.
- Kyun karte hain?* Taaki hum record kar sakein ki image kaise bigadti hai.

Step 2: Reverse Diffusion (Creation - Wapas Banana)

- Yahi asli jaadu hai. Hum ek Neural Network (aksar U-Net) ko train karte hain.
- Hum use wo "Pure Noise" dikhate hain aur kehte hain: *"Isme se thoda sa noise hatao aur batao piche kya image chupi hai."*
- AI guess karta hai aur thoda noise hatata hai.
- Ye process 50 se 100 baar (steps) repeat hota hai.
- Har step par image thodi saaf hoti jati hai, aur ant mein ek HD Image ban jati hai.

18

Multimodal AI

Ab tak humne dekha ki AI ya toh Text samajhta tha (LLMs), ya Images banata tha (Diffusion), ya Audio sunta tha. Lekin **Multimodal AI** in sabka **All-in-One Combo** hai.

Simple shabdon mein:

"Multimodal AI wo hai jo dekh sakta hai (Vision), sun sakta hai (Audio), aur padh sakta hai (Text) — sab ek saath."

Jaise hum insaan baat karte waqt sirf shabdon par dhyan nahi dete, balki saamne wale ke *chehre ke expressions* (Image) aur *aawaz ke tone* (Audio) ko bhi samajhte hain, Multimodal AI bhi wahi karta hai.

19

Vision Language Model

VLM Kya Hai? (The Core Concept)

VLM do badi technologies ka mishran (mixture) hai:

1.Vision Encoder (Aankhein): Ye image ko scan karta hai aur uske features nikalta hai (e.g., "Yahan ek billi hai", "Rang safed hai").

2.Language Model (Dimaag/Zubaan): Ye un features ko samajhta hai aur insaani bhasha mein convert karta hai.

Result: Aap ek photo upload karte hain aur puchte hain *"Is photo mein kya gadbad hai?"* aur VLM uska jawab text mein deta hai.

20

Synthetic Data

Synthetic Data ka matlab hai wo data jo **asli duniya (real world)** se collect nahi kiya gaya, balki **computer programs aur AI algorithms** dwara banaya gaya hai.

Agar simple shabdon mein kahein:

"Ye nakli data hai jo bilkul asli data jaisa behave karta hai, taaki AI ko train kiya ja sake bina kisi ki privacy khatre mein daale."

20

Synthetic Data

Simple Analogy: Video Game vs. Real Driving

Imagine karo aapko gaadi chalana seekhna hai.

1.Real Data: Aap ek asli car lekar road par nikal jao.

(Risk: Accident ho sakta hai, petrol ka kharcha, traffic ki problem).

2.Synthetic Data: Aap ek **High-Quality Video Game** (simulator) mein gaadi chalao.

1. Wahan sadak, gaadi, aur traffic sab computer ne banaya hai (**Synthetic**).
2. Lekin rules wahi hain (break lagane par gaadi rukti hai).
3. Aap bina kisi risk ke 100 baar crash kar sakte ho aur seekh sakte ho.

AI ki training mein ye "Video Game wala data" hi **Synthetic Data** kehlata hai.

AI KNOWLEDGE HIERARCHY & EVOLUTION



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- 11. AutoML
- 12. Model Compression (Pruning & Quantization)

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4

LEVEL 4: INFORMATION RETRIEVAL & KNOWLEDGE SYSTEMS

- Vector Database
- RAG (Retrieval Augmented Generation)
- Knowledge Graphs
- Explainable AI (XAI)
- Federated Learning

LEVEL 4 :Information Retrieval & Knowledge system

Advanced +



21 Vector Database

Vector Database wo jagah hai jahan AI apni "Yaadein" (Memories) store karta hai.

Traditional databases (jaise SQL/Excel) sirf **Exact Match** samajhte hain. Lekin AI ko **Meaning (Matlab)** samajhna hota hai. Vector Database wahi technology hai jo AI ko "Context" aur "Meaning" ke basis par data dhundhna sikhati hai.

Simple shabdon mein:

"Agar **SQL Database** ek **Dictionary** hai (jahan exact word dhundhna padta hai), toh **Vector Database** ek **librarian** hai (jise aap story samjhaoge toh wo waisi hi kitab nikal kar de dega)

LEVEL 4 :Information Retrieval & Knowledge system Advanced +

22

RAG Retrieval Augmented Generation

RAG (Retrieval-Augmented Generation) wo technique hai jo ChatGPT jaise models ko "Open Book Exam" dena sikhata hai.

Agar simple shabdon mein kahein:

"LLM (dimaag) ke paas apni yaaddasht hai, lekin RAG use ek 'Kitaab' (External Data) pakda deta hai taaki wo padh kar sahi jawab de sake, na ki sirf yaad karke."

Bina RAG ke AI "Ratta maar ke" exam deta hai. RAG ke saath wo "Kitaab khol kar" answer dhundhta hai.

LEVEL 4 :Information Retrieval & Knowledge system Advanced +

22

RAG Retrieval Augmented Generation

Problem: LLMs ki Kamiyan

RAG kyun chahiye? Kyunki GPT-4 ya Llama jaise models mein 2 badi problems hain:

1.Knowledge Cutoff (Purana Data):

1. Agar model ki training 2023 mein khatam hui hai, toh use aaj ki news nahi pata.

2.Hallucination (Fenkna):

1. Agar use answer nahi pata, toh wo confidence ke saath jhoot bol deta hai.

3.Private Data:

1. ChatGPT ko aapki company ki internal policies ya emails ke baare mein kuch nahi pata.

Solution: RAG in problems ko solve karta hai AI ko **Live Data** ya **Private Data** ka access dekar.

LEVEL 4 :Information Retrieval & Knowledge system Advanced +

22

RAG Retrieval Augmented Generation

RAG Kaam Kaise Karta Hai? (The 3 Steps)

RAG ka full form hi iska process batata hai:

1.R - Retrieval (Dhundhna):

1. Jab user sawal puchta hai, system sabse pehle aapke database (Vector DB) mein search karta hai aur relevant information nikalta hai.

2.A - Augmentation (Jodna):

1. Us nikali hui information ko user ke sawal ke saath jod diya jata hai (Context banana).

3.G - Generation (Jawab Banana):

1. Ab ye poora masala (Sawal + Information) LLM ko diya jata hai. LLM ise padhta hai aur answer generate karta hai.



LEVEL 4 :Information Retrieval & Knowledge system

Advanced +

23

Knowledge Graphs

Knowledge Graph AI ka wo "Dimaag" hai jo data ko sirf store nahi karta, balki **connect** karta hai aur uska **matlab (meaning)** samajhta hai. Agar simple shabdon mein kahein:

"Agar **SQL Database** ek **Excel sheet** hai (**Rows & Columns**), toh **Knowledge Graph** ek **Detective ka Board** hai jahan **photos (data)** ek dusre se laal dhaage (**relationships**) se judi hoti hain."

Ye AI ko insaanon ki tarah "**Rishte**" (**Relationships**) aur "**Context**" samajhne mein madad karta hai.

LEVEL 4 :Information Retrieval & Knowledge system

Advanced +

24

Explainable ai (XAI)

Explainable AI (XAI) wo technique hai jo humein ye samajhne mein madad karti hai ki **AI ne koi faisla kyun liya.**

Aajkal AI models (jaise Deep Learning) itne complex ho gaye hain ki wo sahi jawab toh de dete hain, par **kaise diya** ye batana mushkil hota hai. Isi problem ko solve karne ke liye **XAI** aaya hai.

Simple shabdon mein:

"Normal AI ek Student ki tarah hai jo bas 'Answer' likh deta hai. Explainable AI wo Student hai jo Answer ke saath 'Steps aur Logic' bhi dikhata hai."

LEVEL 4 :Information Retrieval & Knowledge system

Advanced +

25

Federated Learning

Federated Learning (FL) AI ki duniya mein **Privacy** ka sabse bada rakshak (guardian) hai.

Ab tak humne jitne bhi AI models dekhe (LLMs, Vision, etc.), wo sab **"Centralized"** tareeke se kaam karte hain: "Saara data user se lo aur ek bade server par jama karo, phir wahan model train karo."

Lekin Federated Learning ka concept ulta hai:

"Data ko Model ke paas mat laao, balki Model ko Data ke paas le jao."

Simple shabdon mein: **Aapka personal data aapke phone se bahar kabhi nahi jata, phir bhi AI seekh leta hai.**

AI KNOWLEDGE HIERARCHY & EVOLUTION



LEVEL 5: AUTONOMOUS AI & FUTURE TECH (Cutting-Edge)

- 26. AI Agents
- 27. Edge AI
- 28. Zero-Shot Learning
- 29. AGI (Artificial General Intelligence)
- 30. ASI (Artificial Super Intelligence)



LEVEL 4: INFORMATION RETRIEVAL & KNOWLEDGE SYSTEMS (Advanced+)

- 21. Vector Database
- 22. RAG (Retrieval Augmented Generation)
- 23. Knowledge Graphs
- 24. Explainable AI (XAI)
- 25. Federated Learning



LEVEL 3: AI MODELING & GENERATIVE AI (Advanced)

- 13. LLM (Large Language Model)
- 14. Fine-Tuning
- 15. LORA (Low-Rank Adaptation)
- 16. GAN (Generative Adversarial Network)
- 17. Diffusion Models
- 18. Multimodal AI
- 19. Vision-Language Models (VLMs)
- 20. Synthetic Data



LEVEL 2: MACHINE LEARNING & DEEP LEARNING CONCEPTS (Intermediate)

- 6. CNN (Convolutional Neural Network)
- 7. RNN (Recurrent Neural Network)
- 8. Transformers
- 9. Self-Supervised Learning
- 10. Reinforcement Learning (RL)
- 11. AutoML
- 12. Model Compression (Pruning & Quantization)

LEVEL 1: AI FOUNDATIONS (Basics)

- 1. Tokenization
- 2. Embeddings
- 3. Inference
- 4. Neural Network
- 5. MCP (Multi-Layer Perceptron)



What You Will Learn in This Course

5

LEVEL 5: AUTONOMOUS AI & FUTURE TECH (Cutting-Edge)

- AI Agent
- Edge AI
- Zero-Shot Learning
- AGI (Artificial General Intelligence)
- ASI (Artificial Super Intelligence)

26

AI Agent

AI Agent Artificial Intelligence ka wo "Action Mode" hai jo sirf baatein nahi karta, balki **kaam (tasks)** karke deta hai.

Ab tak humne **LLMs (ChatGPT)** dekha, jo ek "Smart Advisor" ki tarah hai. Aap sawal puchte ho, wo jawab deta hai.

Lekin **AI Agent** ek "Smart Employee/Intern" ki tarah hai. Aap use goal dete ho, aur wo khud tools use karke us goal ko pura karta hai.



26

AI Agent

AI Agent Kya Hai? (The Core Concept)

AI Agent ek aisa system hai jiske paas 3 cheezein hoti hain:

- 1.**Brain (LLM):** Sochne ke liye (jaise GPT-4).
- 2.**Tools (Haath-Pair):** Kaam karne ke liye (Web Search, Calculator, Code Editor, APIs).
- 3.**Agency (Autonomy):** Khud decision lene ki taqat.

Difference Example:

- Normal AI:** Aapne pucha "*Mumbai ka weather kaisa hai?*" -> Wo apne purane data se jawab dega (shayad galat ho).
- AI Agent:** Aapne pucha "*Mumbai ka weather kaisa hai?*" -> Ye turant **Google Search Tool** use karega, aaj ka weather check karega, aur sahi jawab dega.

26

AI Agent

AI Agent ek loop mein kaam karta hai jise aksar "Perception-Action Loop" kehte hain.

1.Goal (Maalik ka hukum): User ne bola: *"Mere liye ek website banao aur host kar do."*

2.Planning (Sochna):

1. Agent ka dimaag (LLM) sochega: *"Pehle code likhna padega, phir HTML file banani padegi, phir server par upload karna padega."*

3.Tool Use (Action):

1. Wo Code Interpreter tool use karke code likhega.
2. Agar error aaya, toh wo khud debug karega.

4.Observation (Feedback):

1. Wo check karega ki code chala ya nahi.

5.Completion:

1. Jab kaam ho jayega, wo aapko boleگا: *"Sir, website live hai, ye raha link."*

Is process mein insaan ki zarurat nahi padi, agent ne khud steps liye.

LEVEL 5 :Autonomous AI & Future Tech – Cutting Edge

27

Edge AI

Edge AI ka simple matlab hai: "**AI jo Internet (Cloud) par nahi, balki aapke Device (Phone, Watch, Car) par chalta hai.**"

Ab tak humne dekha ki zyadatar AI (jaise ChatGPT) **Cloud** par chalta hai. Aap sawal bhejte ho, wo server par jata hai, process hota hai, aur jawab wapas aata hai. Lekin **Edge AI** mein ye processing servers par nahi, balki "**Edge**" (kinare) par hoti hai—yaani jahan data paida ho raha hai (aapke haath mein).

Simple shabdon mein:

"**Agar Cloud AI 'Restaurant se khana mangwana' hai (Time lagta hai), toh Edge AI 'Ghar par khana banana' hai (Turant mil jata hai).**"

28

Zero Shot Learning

Zero-Shot Learning (ZSL) AI ki wo capability hai jahan model un cheezon ko pehchan leta hai ya un tasks ko kar leta hai **jo usne training ke dauran kabhi dekhe hi nahi.**

Normal AI (Supervised Learning) ko sikhana padta hai: "Ye billi hai, ye kutta hai." Tab wo inhe pehchanta hai. Lekin **Zero-Shot Learning** mein AI bina kisi example ke, sirf **Description** ya **Logic** ke dum par nayi cheezon ko pehchan leta hai.

Simple shabdon mein:

"Ye waisa hi hai jaise ek bachche ko 'Zebra' ki photo na dikhayi jaye, bas bataya jaye ki 'Ye ghode (Horse) jaisa dikhta hai par uspe kaali-safed dhariyan (stripes) hoti hain.' Jab bachcha pehli baar Zebra dekhega, wo use pehchan lega."

29

Artificial General Intelligence AGI

Artificial General Intelligence (AGI) AI ki wo future stage hai jahan machine ka dimaag bilkul insaan ke dimaag jaisa ho jayega.

Isse simple points mein samjhein:

1.Kya hai AGI?

1. Abhi ke AI (Narrow AI) sirf **ek kaam** mein expert hote hain (jaise ChatGPT likhne mein, AlphaGo chess mein).
2. **AGI** wo AI hoga jo **koi bhi kaam** kar sakega—coding, painting, gaadi chalana, nayi bhasha seekhna, aur common sense lagana—bina kisi insaan ki madad ke.

2.Difference:

1. **Narrow AI:** Ek calculator jaisa hai (sirf hisaab kar sakta hai).
2. **AGI:** Ek student jaisa hoga (jo maths, science, sports sab seekh sakta hai).

3.Current Status:

1. Abhi tak AGI **bana nahi hai**. Ye sirf theory aur movies (jaise Iron Man ka **JARVIS**) mein hai. Scientists maante hain ki ye aane wale kuch dashakon (decades) mein aa sakta hai.

Short mein: AGI matlab ek aisa computer jo **sochne, samajhne aur seekhne** mein insaan se kam na ho.

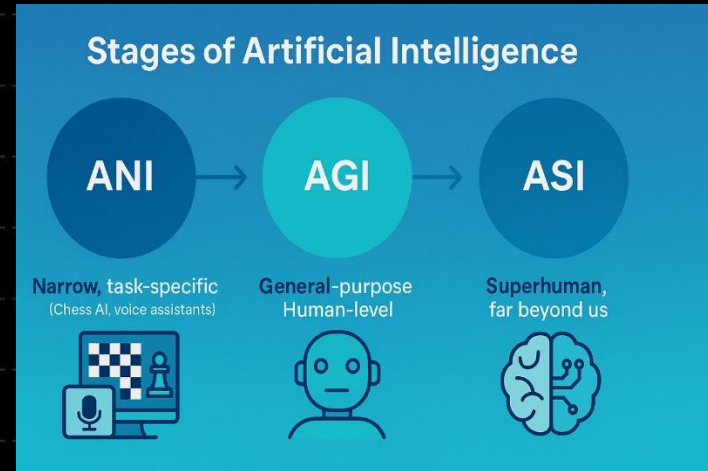
30

Artificial Super Intelligence ASI

Artificial Super Intelligence (ASI) AI ki wo **"God-like"** stage hai jahan machine ki intelligence insaani dimaag se **kahin zyada aage** nikal jati hai.

Agar AGI (General Intelligence) insaan ka **Barabari (Equal)** hai, toh ASI insaan ka **Baap (Superior)** hai.
Simple shabdon mein:

"ASI ek aisa dimaag hai jo duniya ke sabse intelligent scientists (jaise Einstein) se bhi karodon guna tez, creative, aur samajhdaar hoga. Ye wo problems chutki mein solve kar dega jo insaanon ko solve karne mein hazaron saal lagenge."



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